

Supplementary material: the derivations

to “Universe24: a simple integer rule and the group of order 24 give the two slits, Bell’s correlation and the weak-field forms of gravity as clicks”

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This supplement gives the algebraic steps of every result the paper marks DERIVED or COMPUTED without a proof in its text, in the order of the paper’s sections; the theorems proved in the paper are not repeated. Each derivation names its inputs from the paper, its steps, and, where a number was computed, the computation and its check. The notation is the paper’s: Rule3’s coefficients at a Node of clock pace p_0 and Link paces p_a are $R_a = 2 \text{ num } p_a^2$ on the two arrivals of the axis a , $S = 12 \text{ den } \Gamma^2 - 12(\text{den} - \text{num}) p_0^2 - 4 \text{ num } \sum_a p_a^2$ on the Node’s own level, and the wall $w = 6 \text{ den } \Gamma^2$; the content read at the Node is $x = c/\Gamma$, also written U ; $g = (\text{den} - \text{num})/\text{den} = 1 - \cos \omega_0$.

The model (Section 3 of the paper)

Derivation S.1 (the band at a pace, Eq. (4) of the paper). Inputs: *Rule3’s line*, $w a_{\text{next}} = \sum_a R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}}$, with the remainders dropped (they are bounded by w and do not move a plane wave’s rotation). Steps: for a plane wave $a_t(\mathbf{x}) = A \cos(\mathbf{k} \cdot \mathbf{x} - \omega t)$ the two arrivals of the axis a sum to $2 \cos k_a a_{\text{now}}$ and $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega a_{\text{now}}$, so

$$2 \cos \omega = \frac{2 \sum_a R_a \cos k_a + S}{w} = \frac{4 \text{ num } \sum_a p_a^2 \cos k_a + 12 \text{ den } \Gamma^2 - 12(\text{den} - \text{num}) p_0^2 - 4 \text{ num } \sum_a p_a^2}{6 \text{ den } \Gamma^2},$$

that is $\cos \omega = 1 - g(p_0/\Gamma)^2 - \frac{\text{num}}{3 \text{ den}} \sum_a (p_a/\Gamma)^2 (1 - \cos k_a)$, which is Eq. (4). At the vacuum’s pace, $p_0 = p_a = \Gamma$, this is $\cos \omega = \text{num}/\text{den} - \frac{\text{num}}{3 \text{ den}} \sum_a (1 - \cos k_a) = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$; at $k = 0$, $\cos \omega_0 = \text{num}/\text{den}$. Light’s speed and dispersion: for $[1, 1]$ along one axis $\cos \omega = (2 + \cos k)/3$; expanding, $1 - \omega^2/2 + \omega^4/24 = 1 - k^2/6 + k^4/72$, so $\omega = (k/\sqrt{3})(1 - k^2/36 + \dots)$ and the group velocity $d\omega/dk = \sin k/(3 \sin \omega) = (1/\sqrt{3})(1 - k^2/12 + \dots)$: the speed $1/\sqrt{3}$ at long wavelength and the group speed falling as $1 - 0.083 k^2$. Check: at $k = 0.05, 0.1$ and 0.2 the exact $\sqrt{3} d\omega/dk - 1$ over k^2 is $-0.0833, -0.0834$ and -0.0835 .

Derivation S.2 (light’s index at a content, Section 3.4). Inputs: Eq. (4) with the paces of Eq. (3), $p_0^2 = \Gamma^2(1 - 2x + 2x^2)$ and $p_a = \Gamma(1 - 2x)$ for a reader at the weight 1 and no tension. Steps: for light, $g = 0$, so $\cos \omega = 1 - \frac{1}{3}(1 - 2x)^2 \sum_a (1 - \cos k_a)$, and at long wavelength $\omega = (1 - 2x)|\mathbf{k}|/\sqrt{3}$. Light’s speed at the content x is $(1 - 2x)$ times its vacuum speed, so its index is $n = 1/(1 - 2x)$ and $n - 1 = 2x + O(x^2)$: the Link’s pace reads the content twice, and so does light.

Derivation S.3 (the guard’s edge and the horizon, Section 3.4). Inputs: Eq. (4). Steps: the lowest rotation of the band is at $k_a = \pi$ on every axis, where $2 \cos \omega = 2 - 2g(p_0/\Gamma)^2 - \frac{4 \text{ num}}{3 \text{ den}} \sum_a (p_a/\Gamma)^2$; the recurrence $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega a_{\text{now}}$ is bounded iff $|2 \cos \omega| \leq 2$. With $p_0 = p_a = p$ the condition $2 \cos \omega \geq -2$ reads $(p/\Gamma)^2 (2g + 4 \text{ num}/\text{den}) \leq 4$, that is $(p/\Gamma)^2 \leq 2 \text{ den}/(\text{den} + \text{num})$, so the edge is $P^2 = 2 \text{ den } \Gamma^2 \text{ div } (\text{den} + \text{num})$, which is Γ^2 for light. The horizon: $R_a = 2 \text{ num } p_a^2$ vanishes where $p_a = \Gamma(1 - 2x) = 0$, at $x = 1/2$. The clock’s pace there is $(p_0/\Gamma)^2 = (1 - x)^2 + x^2$, whose minimum over x is $1/2$ at $x = 1/2$: the clock never stops, and at the horizon it runs at $1/\sqrt{2}$ of the vacuum’s rate. The frozen Node: with all three $p_a = 0$ the line is $a_{\text{next}} + a_{\text{before}} = (S/w) a_{\text{now}}$ with $S = 12 \text{ den } \Gamma^2 - 12(\text{den} - \text{num}) \Gamma^2/2 = 6 \Gamma^2 (\text{den} + \text{num})$, so $2 \cos \omega = (\text{den} + \text{num})/\text{den} = 1 + \cos \omega_0$; $5/3$ at $[2, 3]$, a period of 10.7 intervals.

Derivation S.4 (the group of order 24 and its action, Section 3.1). Inputs: Rule3’s line, whose six arrivals enter through the three axis coefficients $R_a = 2 \text{ num } p_a^2$ with no coefficient naming an axis beyond its own pace;

the readings D , F and W of Eq. (1). The group: a map of the six Ports onto themselves that keeps opposite Ports opposite is a signed permutation of the three axes, $\mathbf{x} \mapsto \mathbf{P}\mathbf{x}$ with $\mathbf{P}$ a permutation matrix with signs; there are $3! \cdot 2^3 = 48$ of them, the group O_h of the cube, of which the 24 with $\det \mathbf{P} = +1$ are the rotations and the other 24 reflections and rotatory reflections; with the translations of the lattice they form the space group of the simple cubic lattice. Covariance: let σ be such a map and let the turned world carry the levels $a_{\sigma(i)}^\sigma = a_i$ and the paces $p_{\sigma(a)}^\sigma(\sigma(i)) = p_a(i)$. The arrivals of the axis $\sigma(a)$ at $\sigma(i)$ in the turned world are the arrivals of the axis a at i in the original, in either order, and R_a is even in the axis's sign; so the sum $\sum_a R_a(\text{arr}_{+a} + \text{arr}_{-a})$ is the same number at $\sigma(i)$ as at i , S and w are the same, and the division gives the same quotient and remainder: the step commutes with σ , and a world turned by σ runs as the turned world. A translation commutes for the same reason. The readings' kinds: D_i is one number at a Node with no axis in it, carried to $\sigma(i)$ unchanged, a scalar; the currents F_{ij} are attached to the Links and carried with them, and the momentum density $P_a(i) = (F_{i,+a} - F_{i,-a})/\text{num}$ changes sign where σ reverses the axis a and is carried to the image axis otherwise, a vector; W_i carries no axis and is carried unchanged, and it changes sign under the reflection of a plane record's sense, $\text{im} \mapsto -\text{im}$, a pseudoscalar in that sense; the click (R, t) names a region and an interval and is carried to the turned region, a scalar. The time reversal: exchanging now and before leaves $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$ unchanged once next and before exchange their roles, since D is symmetric in the pair (next, before); it reverses F_{ij} and W_i , which are antisymmetric in (now, before); and it carries the step to its inverse by Theorem 1. The rational rotation: the two parts of a plane record obey the same linear line with the same coefficients, so any linear combination of them does; the combination $(\text{re}', \text{im}') = ((\text{pre} - q \text{im})/r, (q \text{re} + p \text{im})/r)$ with $p^2 + q^2 = r^2$ carries solutions to solutions, and its Wronskian is $(p^2 + q^2)/r^2$ times the old one, unchanged. What is not in the group: a boost changes the record's wave number and rotation together, which no permutation of Ports does, and a rotation by an angle other than the cube's carries a Node off the lattice; so neither is a symmetry, and Lorentz's factor, where the paper finds it, is a property of the clicks (Derivation S.21).

Exact properties (Section 4)

Derivation S.5 (the share's change is the currents, Eq. (10)). Inputs: the share at the vacuum's paces, $e_i = 3 \text{den}(\text{now}_i^2 + \text{before}_i^2) - \text{num} \text{now}_i S_6(\text{before})_i$, with S_6 the six-neighbour sum; Rule3 at the vacuum's paces, where $S = 0$, $R_a = 2 \text{num} \Gamma^2$ and $w = 6 \text{den} \Gamma^2$, so that $3 \text{den}(\text{next}_i + \text{before}_i) = \text{num} S_6(\text{now})_i + \rho_i$ with $\rho_i = (r_i - r'_i)/\Gamma^2$ the remainders' term. Steps: after the step the pair (now, before) becomes (next, now), so

$$e'_i - e_i = 3 \text{den}(\text{next}_i^2 - \text{before}_i^2) - \text{num}[\text{next}_i S_6(\text{now})_i - \text{now}_i S_6(\text{before})_i].$$

With $3 \text{den}(\text{next}_i + \text{before}_i) = \text{num} S_6(\text{now})_i + \rho_i$ the first term is $(\text{next}_i - \text{before}_i)(\text{num} S_6(\text{now})_i + \rho_i)$, and the bracket cancels its next_i part:

$$\begin{aligned} e'_i - e_i &= \text{num}[\text{now}_i S_6(\text{before})_i - \text{before}_i S_6(\text{now})_i] + (\text{next}_i - \text{before}_i) \rho_i \\ &= \sum_{j \sim i} \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j) + (\text{next}_i - \text{before}_i) \rho_i, \end{aligned}$$

the six currents F_{ij} into the Node plus the remainder term. Where the three Link paces are equal but not Γ the Node term carries the weight $3\Gamma^2/\sum_a p_a^2$ and the same substitution goes through with $S \neq 0$, the S term cancelling between $\text{next}_i^2 - \text{before}_i^2$ and the bracket as in the proof of Theorem 3; where they differ, the term that does not cancel is the anisotropy term of Eq. (10), $\text{num} \text{now}_i \sum_a (1 - 3p_a^2/P^2) \text{arr}_a$ with $P^2 = \sum_a p_a^2$.

Derivation S.6 (the total share is non-negative, Section 4.4). Inputs: e_i as above on a closed or periodic GameBoard; S_6 symmetric with eigenvalues $2 \sum_a \cos k_a \in [-6, 6]$. Steps: $\sum_i e_i = 3 \text{den}(|\mathbf{n}|^2 + |\mathbf{b}|^2) - \text{num} \mathbf{n} \cdot S_6 \mathbf{b}$, and $|\mathbf{n} \cdot S_6 \mathbf{b}| \leq \|S_6\| |\mathbf{n}| |\mathbf{b}| \leq 6 |\mathbf{n}| |\mathbf{b}| \leq 3(|\mathbf{n}|^2 + |\mathbf{b}|^2)$, so $\sum_i e_i \geq 3(\text{den} - |\text{num}|)(|\mathbf{n}|^2 + |\mathbf{b}|^2) \geq 0$ for $|\text{num}| \leq \text{den}$. Equality needs $|\text{num}| = \text{den}$, $\mathbf{b}$ an eigenvector of S_6 at 6 (the uniform level) and $\mathbf{n} = \mathbf{b}$: a static uniform level of light, whose share is 0. A Node's share: at fixed before_i and $S_6(\text{before})_i = s$, e_i is a quadratic in now_i with the minimum $3 \text{den} \text{before}_i^2 - \text{num}^2 s^2 / (12 \text{den})$, negative iff $|s| > 6(\text{den}/\text{num}) |\text{before}_i|$; a plane record has $|s|/|\text{before}_i| = 2 \sum_a \cos k_a \leq 6$ and is never negative, a lay's edge can be. A region: pairing each boundary Link's cross term and bounding $2|\text{now}_i \text{before}_j| \leq \text{now}_i^2 + \text{before}_j^2$, $\sum_{i \in R} e_i \geq 3(\text{den} - \text{num}) \sum_R (\text{now}_i^2 + \text{before}_i^2) - \frac{\text{num}}{2} \sum_{\text{boundary Ports}} (\text{now}_j^2 + \text{before}_j^2)$ over the outside neighbours j : a region's share is negative at most by half a level-square per boundary Port in the current's units.

Derivation S.7 (the tension of a plane wave, Section 4.3). Inputs: *the stress reading* $G_{aa}(i) = \text{now}_i(\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a}(\text{now}_{i+a} - \text{now}_{i-a})$. Steps: *for* $\text{now}_i = b \cos(kx_i + \phi)$ *along the axis*, with $c_n = \cos(kx_i + nk + \phi)$, $G = b^2[c_0c_2 - c_0^2 - c_1^2 + c_1c_{-1}]$; *the identities* $c_0c_2 = \frac{1}{2}[\cos(2kx + 2k + 2\phi) + \cos 2k]$, $c_1c_{-1} = \frac{1}{2}[\cos(2kx + 2\phi) + \cos 2k]$, $c_0^2 = \frac{1}{2}[1 + \cos(2kx + 2\phi)]$ *and* $c_1^2 = \frac{1}{2}[1 + \cos(2kx + 2k + 2\phi)]$ *cancel the phase-dependent parts*, leaving $G = b^2(\cos 2k - 1) = -2b^2 \sin^2 k$ *at every Node and interval*; *so* $T_{aa} = -\text{num}(G(i) + G(i-a)) \text{div } 2 = 2 \text{num } b^2 \sin^2 k$, *and for a wave with the components* k_a, k_b *the cross reading is* $G_{ab} = -2b^2 \sin k_a \sin k_b$.

Measurement (Section 5)

Derivation S.8 (the region's reading factor and the draw's scatter, Section 5.3). Steps: *a region of* n *consecutive rows sums a fringe* $\cos(2\pi y/\Lambda)$ *to* $\text{Re} \sum_{y=0}^{n-1} e^{2\pi i y/\Lambda} = \text{Re} \frac{1 - e^{2\pi i n/\Lambda}}{1 - e^{2\pi i/\Lambda}}$, *whose modulus is* $\sin(\pi n/\Lambda)/\sin(\pi/\Lambda)$; *over* n *rows the fringe's modulation is kept with the factor* $\sin(\pi n/\Lambda)/(n \sin(\pi/\Lambda))$, 0.90 *at* $n = 4$, $\Lambda = 16$, *and* 0 *at* $\Lambda = n$. *A frequency read as a click rate over* N *intervals resolves* $\Delta\omega \geq 2\pi/N$, *the window's own resolution.* N *quanta drawn independently by the shares* p_i *give each region a binomial count*, $Np_i \pm \sqrt{Np_i(1-p_i)}$.

Derivation S.9 (the Wronskian's conservation and its current, Section 5.5). Inputs: *the two parts of a plane record step by the same symmetric read matrix* $\mathbf{M}$ *(the paper's Theorem 3)*, $\mathbf{re}_{\text{next}} = \mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}}$ *and the same for* $\mathbf{im}$, *with the weights* $\mathbf{D}$ *of the conserved form*, $\mathbf{D}\mathbf{M}$ *symmetric*. Steps: *let* $W = \mathbf{re}_{\text{now}}^\top \mathbf{D} \mathbf{im}_{\text{before}} - \mathbf{im}_{\text{now}}^\top \mathbf{D} \mathbf{re}_{\text{before}}$. *After the step it is* $\mathbf{re}_{\text{next}}^\top \mathbf{D} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{next}}^\top \mathbf{D} \mathbf{re}_{\text{now}} = (\mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}})^\top \mathbf{D} \mathbf{im}_{\text{now}} - (\mathbf{M} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{before}})^\top \mathbf{D} \mathbf{re}_{\text{now}}$; *the* $\mathbf{M}$ *terms are* $\mathbf{re}_{\text{now}}^\top \mathbf{M}^\top \mathbf{D} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{now}}^\top \mathbf{M}^\top \mathbf{D} \mathbf{re}_{\text{now}}$, *which vanish since* $\mathbf{M}^\top \mathbf{D} = \mathbf{D}\mathbf{M}$ *is symmetric*, *and the rest is* W *again*. *Per Node the same cancellation leaves the Link terms*, $\text{num}(\mathbf{im}_i \mathbf{re}_j - \mathbf{re}_i \mathbf{im}_j)$, *the Wronskian's current*. *For the circular record* $\mathbf{re} = A \cos \omega t$, $\mathbf{im} = eA \sin \omega t$ *at one Node*, $W = A \cos \omega t \cdot eA \sin \omega(t-1) - eA \sin \omega t \cdot A \cos \omega(t-1) = -eA^2 \sin \omega$, *constant*.

Derivation S.10 (Bell's correlation under the declared credit, Section 5.7 and Section 9.4). Inputs: *two parts with the products* M_1, M_2 *of their amplitudes at the two sides' ports*; *the port coefficients* $e(+)= (\cos a, \sin a)$, $e(-)= (-\sin a, \cos a)$ *on side* A *and the same with* b *on side* B ; *the joint share* $J(p, q) = (\sum_k e_k(p) e_k(q) M_k)^2$; $E = \sum_{p,q} s(p)s(q) J(p, q) / \sum_{p,q} J(p, q)$ *with* $s(\pm) = \pm 1$. Steps: *write* $c_a = \cos a$, $s_a = \sin a$ *and the same for* b . *The four sums are* $M_1 c_a c_b + M_2 s_a s_b$, $M_1 s_a s_b + M_2 c_a c_b$, $-M_1 c_a s_b + M_2 s_a c_b$ *and* $-M_1 s_a c_b + M_2 c_a s_b$ *for* $(+, +)$, $(-, -)$, $(+, -)$ *and* $(-, +)$. *Their squares sum to* $(M_1^2 + M_2^2)(c_a^2 c_b^2 + s_a^2 s_b^2 + c_a^2 s_b^2 + s_a^2 c_b^2) = M_1^2 + M_2^2$, *the cross terms cancelling in pairs*. *The signed sum is* $(M_1^2 + M_2^2)(c_a^2 - s_a^2)(c_b^2 - s_b^2) + 8M_1 M_2 c_a s_a c_b s_b$, *so*

$$E(a, b) = \cos 2a \cos 2b + \rho \sin 2a \sin 2b, \quad \rho = \frac{2M_1 M_2}{M_1^2 + M_2^2},$$

and the marginal $P(A+) = [(M_1 c_a c_b + M_2 s_a s_b)^2 + (-M_1 c_a s_b + M_2 s_a c_b)^2] / (M_1^2 + M_2^2) = (M_1^2 c_a^2 + M_2^2 s_a^2) / (M_1^2 + M_2^2)$, *independent of* b . *For equal parts* $\rho = 1$ *and* $E = \cos 2(a - b)$. *The integer settings: with* $e(+)= (p, q)$ *and* $e(-)= (-q, p)$ *unnormalised and equal parts*, $J(+, +) = J(-, -) = (p_{APB} + q_{AQB})^2$ *and* $J(+, -) = J(-, +) = (p_{AQB} - q_{APB})^2$, *whose sum is* $2(p_A^2 + q_A^2)(p_B^2 + q_B^2)$ *by Lagrange's identity*, *so* $E = [(p_{APB} + q_{AQB})^2 - (p_{AQB} - q_{APB})^2] / [(p_A^2 + q_A^2)(p_B^2 + q_B^2)]$, *a rational number*. *At* $a = (1, 0)$, $a' = (1, 1)$, $b = (12, 5)$, $b' = (5, 12)$: $E(a, b) = (144 - 25)/169 = 119/169$, $E(a, b') = (25 - 144)/169 = -119/169$, $E(a', b) = (17^2 - 7^2)/338 = 120/169$, $E(a', b') = 120/169$, *and* $S = E(a, b) - E(a, b') + E(a', b) + E(a', b') = 478/169 = 2.8284$. *At the ideal angles* $S = \sqrt{2}(1 + \rho)$; *at the integer settings* $S = (238 + 240\rho)/169$. *The two local credits: with the pair's phase* θ *uniform*, *a credit of each side by the shares of its ports gives* $E = \langle \cos 2(a - \theta) \cos 2(b - \theta) \rangle_\theta = \frac{1}{2} \cos 2(a - b)$ *and* $S = \sqrt{2}$; *a credit by the sign of the larger port gives* $E = \langle \text{sgn} \cos 2(a - \theta) \text{sgn} \cos 2(b - \theta) \rangle_\theta = 1 - 4|a - b|/\pi$, *the sawtooth, whose sum at the four settings is exactly* 2 . *Check: the averages over* θ *on a grid of* 2×10^5 *points give* $S = 1.4142$ *and* 2.0000 , *and the contrast credit, which discards the pairs near the contrast's zero*, $S = 2.8284$ *at the efficiency* $2/\pi$.

The families (Section 6)

Derivation S.11 (the reach of a field with a gap, Section 6.4). Inputs: *a field's rest outside its source, the line with* $a_{\text{next}} = a_{\text{before}} = a_{\text{now}} = a$: $6 \text{den } a = \text{num } S_6(a)$ *at the vacuum's pace*, *that is* $\text{num}(S_6 - 6)a = 6(\text{den} - \text{num})a$. Steps: *for a rest falling along one axis as* $e^{-\kappa x}$ *and uniform across*, $S_6(a) = (2 \cosh \kappa + 4)a$, *so* $6 \text{den} = \text{num}(2 \cosh \kappa + 4)$ *and* $\cosh \kappa = 3 \text{den}/\text{num} - 2$. *For* num *just under* den , $\cosh \kappa = 1 + \kappa^2/2 + \dots$,

so $\kappa^2 = 6 \text{ den}/\text{num} - 6 = 6(\text{den} - \text{num})/\text{num}$ and the range is $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$ Links: 20.00 at [2400, 2401]; [1, 2] has $\cosh \kappa = 4$, $e^{-\kappa} = 0.127$. The gap is also the rest rotation of the field's own travelling events, $\cos \omega_g = \text{num}/\text{den}$, $\omega_g^2 = 2(\text{den} - \text{num})/\text{den} + \dots$, whose Compton length at light's speed, $c/\omega_g = \sqrt{\text{den}/(6(\text{den} - \text{num}))} = R\sqrt{\text{den}/\text{num}}$, is the range to the gap's first order: Yukawa's identity with no input.

Derivation S.12 (the two speeds, Section 6.5). Steps: light at a content x travels at $(1-2x)/\sqrt{3}$ (Derivation S.2) and the massless field's own events, which read no content, at $1/\sqrt{3}$; over L Links they arrive $L\sqrt{3}[1/(1-2x)-1] = 2xL\sqrt{3} + O(x^2)$ intervals apart at long wavelength. GW170817 bounds the difference of the two speeds at 7×10^{-16} of light's, so $2x < 7 \times 10^{-16}$ and the vacuum content over Γ is below 10^{-15} .

Bodies (Section 7)

Derivation S.13 (the bound body: the fixed point and the functional, Section 7.2). Inputs: Rule3's line at the paces of the wells the record sources; a standing record $a_{i,t} = \phi_i \cos(\omega_b t)$ with one rotation at every Node; the clock's pace $(p_0/\Gamma)^2 = 1 - 2\Phi_i/\Gamma + \dots$ at a well Φ_i (in levels). The fixed point's equation: with the remainders dropped the line gives at every Node

$$2 \cos \omega_b \phi_i = \frac{1}{w} \left[\sum_a R_a(i) (\phi_{i+a} + \phi_{i-a}) + S_i \phi_i \right] = (\mathbf{M}\phi)_i,$$

so ϕ is an eigenvector of the read matrix $\mathbf{M}$ and the standing record is its top mode, the maximiser of $\phi^\top \mathbf{M}\phi$ on the sphere $\sum \phi_i^2 = 1$ (the generator of Section 7.1 iterates this read act). To the first order in Φ/Γ , $S_i/w = 2 - 2g(1 - 2\Phi_i/\Gamma) - \frac{2\text{num}}{3\text{den}} \sum_a (p_a/\Gamma)^2 = 2 \cos \omega_0 + 4g\Phi_i/\Gamma - \frac{2\text{num}}{3\text{den}} \sum_a (p_a/\Gamma)^2$ and $R_a/w = \frac{\text{num}}{3\text{den}} (p_a/\Gamma)^2$, so, keeping the band's term at the vacuum's Link pace (its product with the well is of the second order),

$$(\mathbf{M}\phi)_i = 2 \cos \omega_0 \phi_i + \frac{4g}{\Gamma} \Phi_i \phi_i - \frac{\text{num}}{3\text{den}} (-\Delta^2 \phi)_i, \quad (-\Delta^2 \phi)_i = 6\phi_i - \sum_a (\phi_{i+a} + \phi_{i-a}).$$

The functional: the well is sourced by the record itself, $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$ for a body of M quanta with the counts $c_j = M\phi_j^2$. Let

$$F[\phi] = \frac{\text{num}}{3\text{den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2,$$

the paper's Eq. (12). Its gradient is $\partial F/\partial \phi_i = \frac{2\text{num}}{3\text{den}} (-\Delta^2 \phi)_i - \frac{8g}{\Gamma} \Phi_i \phi_i$, using the symmetry of G_r (each well counted once, Hartree's form, is what makes the factor $2g$ in F give $4g$ in the gradient). So $\partial F/\partial \phi = 2\lambda\phi$ on the sphere is exactly the fixed point's equation with $\lambda = 2 \cos \omega_0 - 2 \cos \omega_b$: a body is a critical point of F on the sphere, and the virial and the second variation along the dilation are Theorem 4 of the paper. The scaling powers: under $\phi_R(\mathbf{x}) = R^{-d/2} \phi(\mathbf{x}/R)$ the sum over Links of $(\phi_i - \phi_j)^2$ is the lattice form of $\int |\nabla \phi|^2$ and scales as R^{-2} , the pressure K/R^2 ; the well energy $\sum_{ij} \phi_i^2 G_{ij} \phi_j^2$ is $\iint \rho(\mathbf{x}) G(\mathbf{x} - \mathbf{y}) \rho(\mathbf{y})$ with $\rho = \phi^2$ of unit mass, which scales as $R^{-(d-2)}$ for the massless kernel $G \sim 1/r$ and for a screened kernel within its reach, and as R^{-d} for a screened kernel beyond its reach, where G acts as a point kernel and the energy is $\int \rho^2$: the powers s_r of Theorem 4. The cloud's size: for the massless kernel in three dimensions $F(R) = a/R^2 - c/R$ with $a = \frac{\text{num}}{3\text{den}} \int |\nabla \phi|^2$ and $c = \frac{2g}{\Gamma} \frac{M}{E} \iint \frac{\rho \rho'}{4\pi|\mathbf{x}-\mathbf{y}|} \cdot 3$ at the unit size, so the minimum is at $R = 2a/c \propto \Gamma E/(gM)$: the radius falls as $1/M$ and grows with Γ , the paper's $2\Gamma/C$ up to the shape's constants, which the generator computes.

Derivation S.14 (the invariant and the count's drift, Eq. (13)). Inputs: the one-Node line $a_{t+1} + a_{t-1} = 2 \cos \omega_t a_t$ with ω_t changing slowly. Steps: at fixed ω the form $D_t = a_t^2 - a_{t+1}a_{t-1} = a_t^2 + a_{t-1}^2 - 2 \cos \omega a_t a_{t-1}$ is conserved and equals $A^2 \sin^2 \omega$ for $a_t = A \cos(\omega t + \phi)$; in the plane (a_t, a_{t-1}) the step is a linear map of determinant 1 whose orbits are the ellipses $D = \text{const}$, of area $\pi D/\sin \omega$. A slow change of ω keeps the area enclosed by the orbit, the action of the map, so $D/\sin \omega = A^2 \sin \omega$ is the adiabatic invariant: the discrete form of E/ω . Equivalently, the ansatz $a_t = C \cos(\sum_{s \leq t} \omega_s + \phi)/\sqrt{\sin \omega_t}$ satisfies the line to the first order in $\omega_{t+1} - \omega_t$. Check: the recurrence stepped 20,000 intervals with ω lowered from 0.841 to 0.600 keeps $D/\sin \omega$ within 4×10^{-5} of its value while D falls by a quarter. The drift: the share at a Node in quanta is $(3\Gamma^2/\sum_a p_a^2) A^2 \sin^2 \omega_b/T$ (Eq. (9) at the Node's paces), so with $A^2 \sin \omega_b$ fixed, $N(t_2)/N(t_1) = [\sin \omega_b(t_2)/\sin \omega_b(t_1)] [\sum_a p_a^2(t_1)/\sum_a p_a^2(t_2)]$, Eq. (13). Planck's relation: the share $A^2 \sin^2 \omega/T$ is the invariant $I = A^2 \sin \omega/T$ times $\sin \omega$.

Derivation S.15 (the clock's click rate, Section 7.4). Inputs: Eq. (4) at $k = 0$ with $p_0^2 = \Gamma^2(1 - 2U + 2U^2)$: $\cos \omega(U) = 1 - g(1 - 2U + 2U^2) = \cos \omega_0 + 2gU - 2gU^2$. Steps: to first order $d(\cos \omega) = 2g dU = -\sin \omega_0 d\omega$, so $d\omega/\omega_0 = -[2g/(\omega_0 \sin \omega_0)] dU = -f(\omega_0) dU$ with $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$; $f = 1.063$ at $[2, 3]$, and as $\omega_0 \rightarrow 0$, $1 - \cos \omega_0 \rightarrow \omega_0^2/2$ and $f \rightarrow 1$, the rate $1 - U$ of the clock's pace alone. The light clock: a packet between two faces L Links apart at the content x has the round trip $2L/v_g = 2L\sqrt{3}/(1 - 2x)$ (Derivation S.2), so its rate falls by $2x$.

Derivation S.16 (the resonance at twice the rotation, Section 7.4). Inputs: the one-Node line with the rotation modulated by a light level $b_L \cos \omega_L t$ read into the content: $2 \cos \omega(t) = 2 - 2g(1 - 2x + 2x^2)$ with $x = x_0 + b_L \cos(\omega_L t)/\Gamma$, so $d(2 \cos \omega)/dx = 4g(1 - 2x)$ and $a_{\text{next}} + a_{\text{before}} = [2 \cos \omega_b + \epsilon \cos(\omega_L t + \vartheta)] a_{\text{now}}$ with $\epsilon = 4g(1 - 2x_0) b_L/\Gamma$. Steps: write $a_t = \text{Re}[z_t e^{i\omega_b t}]$ with z slowly varying, $z_{t \pm 1} = z_t \pm z'$. Then $a_{t+1} + a_{t-1} = \text{Re}[e^{i\omega_b t} (2 \cos \omega_b z_t + 2i \sin \omega_b z')]$, and the product $\epsilon \cos(\omega_L t + \vartheta) \text{Re}[z_t e^{i\omega_b t}]$ has, at $\omega_L = 2\omega_b$, the resonant part $(\epsilon/2) \text{Re}[\bar{z}_t e^{i\vartheta} e^{i\omega_b t}]$. Equating the slowly varying coefficients of $e^{i\omega_b t}$, $2i \sin \omega_b z' = (\epsilon/2) e^{i\vartheta} \bar{z}$, so $z' = -i\epsilon e^{i\vartheta} \bar{z}/(4 \sin \omega_b)$; this antilinear map $z \mapsto \lambda \bar{z}$ with $|\lambda| = \mu$ has the real eigenvalues $\pm\mu$ on two lines of the z plane, so the amplitude grows as $e^{\mu t}$ along one line and the share, quadratic in the amplitude, as $e^{2\mu t}$. Detuned by $\Delta = \omega_L - 2\omega_b$, the substitution $z = \zeta e^{i\Delta t/2}$ gives $\zeta' = -i(\Delta/2)\zeta - i\mu e^{i\vartheta} \bar{\zeta}$, whose exponents are $\pm\sqrt{\mu^2 - \Delta^2/4}$: growth only for $|\Delta| < 2\mu$, a full width $\epsilon/\sin \omega_b$. Check: the recurrence stepped 20,000 intervals at $2 \cos \omega_b = 1.4317$ and $\epsilon = 0.013$ grows at 0.004654 per interval against the formula's 0.004654. The two-mode line: with two bound modes the same modulation multiplies $a_t = \text{Re}[z_i e^{i\omega_i t} + z_j e^{i\omega_j t}]$ and its product with $\cos \omega_L t$ carries the frequencies $\omega_i \pm \omega_L$; it feeds the mode j at $\omega_L = \omega_j - \omega_i$, the line between two bound states, of which $2\omega_b$ is the case $z_i = \bar{z}_j$.

Derivation S.17 (the cross current of two standing records, Section 7.5). Inputs: two standing records $a_t = A \cos(\omega_1 t + \alpha)$ and $b_t = B \cos(\omega_2 t + \beta)$ at the two ends of a Link. Steps: the current $\text{num}(a_t b_{t-1} - a_{t-1} b_t)$ is $\text{num}AB[\cos(\omega_1 t + \alpha) \cos(\omega_2 t - \omega_2 + \beta) - \cos(\omega_1 t - \omega_1 + \alpha) \cos(\omega_2 t + \beta)]$, a sum of cosines of $(\omega_1 - \omega_2)t$ and $(\omega_1 + \omega_2)t$ with the constant parts $\frac{1}{2} \cos(\alpha - \beta + \omega_2)$ and $\frac{1}{2} \cos(\alpha - \beta - \omega_1)$ only when $\omega_1 = \omega_2$; for $\omega_1 \neq \omega_2$ its mean over the beat is 0, so no quantum passes at the first order. The reach: two bodies' tails are $a_1 e^{-\kappa d_1}$ and $a_2 e^{-\kappa d_2}$ at a Link $d = d_1 + d_2$ Links from the two centres, so the current there is of the order $a_1 a_2 e^{-\kappa d}$ per interval and falls under one quantum, T , at $d = \ln(a_1 a_2/T)/\kappa$.

The forms of nature (Section 8)

Derivation S.18 (the fall and Kepler's potential, Eq. (14)). Inputs: Eq. (4) at a content U for a reader at the weight 1, $\cos \omega(k, U) = 1 - g(1 - 2U + 2U^2) - \frac{\text{num}}{3 \text{den}} (1 - 2U)^2 (1 - \cos k)$ along the motion. Steps: at small k and U , $\omega(k, U) = \omega_0(U) + k^2/(2m^*)$ with the inertia $m^* = 3 \text{den} \sin \omega_0 / \text{num} = 3 \tan \omega_0$ (from $\sin \omega_0 d\omega = \frac{\text{num}}{3 \text{den}} k dk$), so the group velocity is $v = k/m^*$. The rays of $\omega(k, x)$ give $dk/dt = -\partial\omega/\partial x = -(d\omega_0/dU) \nabla U = (2g/\sin \omega_0) |\nabla U|$ toward the well (Derivation S.15), so the acceleration is

$$a = \frac{1}{m^*} \frac{dk}{dt} = \frac{\text{num}}{3 \text{den} \sin \omega_0} \cdot \frac{2g}{\sin \omega_0} |\nabla U| = \frac{2 \cos \omega_0 (1 - \cos \omega_0)}{3 \sin^2 \omega_0} |\nabla U| = \frac{2 \cos \omega_0}{3(1 + \cos \omega_0)} |\nabla U|,$$

which in the gap $\epsilon = 1 - \cos \omega_0$ is $[(1 - \epsilon)/(3(1 - \epsilon/2))] |\nabla U|$: 0.2667 at $[2, 3]$ and $1/3$ as the gap closes. Nature's fall is $a = c^2 |\nabla U_K|$ with $U_K = GM_K/(c^2 r)$ and $c^2 = 1/3$ here, so $U_K = [2 \cos \omega_0/(1 + \cos \omega_0)] U$, Eq. (14), 0.800 at $[2, 3]$.

Derivation S.19 (the post-Newtonian parameters against Kepler's potential, Section 8). Inputs: Eq. (14), $U = \lambda U_K$ with $\lambda = (1 + \cos \omega_0)/(2 \cos \omega_0) = (\text{den} + \text{num})/(2 \text{num})$; the three forms in U : the bending $4U_b$, the periapsis $6\pi U_1/(a(1 - e^2))$, the redshift $f(\omega_0) U$. Steps: the bending is $4\lambda U_{K,b} = (2 \text{den}/\text{num}) U_{K,b}$, and the post-Newtonian bending is $(1 + \gamma) 2GM/(c^2 b) = (1 + \gamma) 2U_{K,b}$, so $1 + \gamma_K = 1 + \text{den}/\text{num}$, $\gamma_K = \text{den}/\text{num}$. The periapsis is λ times Einstein's $6\pi U_{K,1}/(a(1 - e^2))$, and the post-Newtonian advance is $(2 - \beta + 2\gamma)/3$ of Einstein's, so $(2 - \beta_K + 2 \text{den}/\text{num})/3 = (\text{den} + \text{num})/(2 \text{num})$, whence $\beta_K = 2 + 2 \text{den}/\text{num} - 3(\text{den} + \text{num})/(2 \text{num}) = (\text{den} + \text{num})/(2 \text{num})$. The redshift is $f(\omega_0) \lambda U_K$, and $f\lambda = \frac{2(1 - \cos \omega_0)}{\omega_0 \sin \omega_0} \cdot \frac{1 + \cos \omega_0}{2 \cos \omega_0} = \frac{\sin^2 \omega_0}{\omega_0 \sin \omega_0 \cos \omega_0} = \frac{\tan \omega_0}{\omega_0}$, so $1 + \alpha_K = \tan \omega_0/\omega_0$. At $[2, 3]$: 1.5, 1.25 and 1.329; all three tend to 1 as $\omega_0 \rightarrow 0$, and $\gamma_K - 1 < 2 \times 10^{-5}$ requires $\text{den}/\text{num} - 1 < 2 \times 10^{-5}$.

Derivation S.20 (Shapiro's delay, Section 8 (c)). Inputs: light's index $n = 1 + 2U$ with $U = \mu/r$ (Derivation S.2). Steps: the delay along a straight path at the impact parameter b from $-x_1$ to x_2 is $\int (n - 1) dl/c =$

$2\sqrt{3}\mu \int_{-x_1}^{x_2} dl/\sqrt{l^2+b^2} = 2\sqrt{3}\mu [\operatorname{arsinh}(x_2/b) + \operatorname{arsinh}(x_1/b)]$, which for $x_1, x_2 \gg b$ is $2\sqrt{3}\mu \ln(4x_1x_2/b^2)$: Shapiro's delay with $\gamma = 1$ in U , and with γ_K against U_K by the same factor as the bending, since the bending is the transverse derivative of the same integral.

Derivation S.21 (the moving clock to the second order, Section 8 (e)). Inputs: a bound band $\omega(k) = \omega_0 + \Delta(1 - \cos k)$; the rotation at the moving centre $\Omega(k) = \omega(k) - k\omega'(k)$; the velocity $v = \omega'(k) = \Delta \sin k$. Steps: $\Omega = \omega_0 + \Delta(1 - \cos k - k \sin k) = \omega_0 - \Delta k^2/2 + \Delta k^4/8 + O(k^6)$, and from $v = \Delta(k - k^3/6 + \dots)$, $k^2 = v^2/\Delta^2 + v^4/(3\Delta^4) + \dots$ and $k^4 = v^4/\Delta^4 + \dots$, so

$$f = \frac{\Omega}{\omega_0} = 1 - \frac{v^2}{2\Delta\omega_0} - \frac{v^4}{24\Delta^3\omega_0} + O(v^6) = 1 - \frac{v^2}{2c_b^2} - \frac{v^4}{24\Delta^3\omega_0} + \dots, \quad c_b^2 = \Delta\omega_0,$$

which agrees with $\sqrt{1 - v^2/c_b^2} = 1 - v^2/(2c_b^2) - v^4/(8c_b^4) - \dots$ at the second order and differs at the fourth, where Lorentz's $-v^4/(8\Delta^2\omega_0^2)$ equals the band's $-v^4/(24\Delta^3\omega_0)$ only at $\Delta = \omega_0/3$. Check: at $\Delta = 0.3$, $\omega_0 = 0.8$, $v = 0.05$ the residual $(f - 1 + v^2/(2c_b^2))/v^4$ is -1.95 against the formula's -1.93 , the difference the sixth order.

Derivation S.22 (de Broglie's relation, Section 8 (h)). Steps: from Derivation S.18, $v = k/m^*$ with $m^* = 3 \tan \omega_0$ at the vacuum's content, so $k = m^*v = \omega_b v/c_m^2$ with $c_m^2 = \omega_0/m^* = \omega_0/(3 \tan \omega_0)$, the family's own light speed; 0.2508 at $[2, 3]$.

The two slits (Section 9)

Derivation S.23 (the fringe's spacing and the near-field minima, Section 9.2). Inputs: the gaps centred at the rows 18 and 30, $d = 12$; the screen $L = 24$ Links from the wall; $\lambda = 8$. Steps: the far-field spacing is $\lambda L/d = 16$ Links. In the near field the minima are where the two gaps' paths to the screen differ by half a wavelength, $\sqrt{L^2 + (y - 18)^2} - \sqrt{L^2 + (y - 30)^2} = \pm 4$, whose solutions are $y = 15.28$ and $y = 32.72$: the rows 15 and 33. Check: the two roots computed from the equation above.

Calibration (Section 10)

Derivation S.24 (the Link's bound from light's dispersion, Section 10.2). Inputs: the group speed $v_g = (1/\sqrt{3})(1 - k^2/12)$ (Derivation S.1); the bound on a quadratic dependence of light's speed on its energy, $|\delta v/v| < 3 \times 10^{-16}$ at 1 TeV. Steps: $k^2/12 < 3 \times 10^{-16}$ gives $k < 6 \times 10^{-8}$ per Link at 1 TeV; with $k = 2\pi \text{Link}/\lambda$ and $\lambda = 1.24 \times 10^{-18}$ m at 1 TeV, the Link is below $k\lambda/(2\pi) = 1.2 \times 10^{-26}$ m.

Derivation S.25 (Newton's constant as a reading, Eq. (18)). Inputs: the massless field's rest far from a static source of s quanta per interval, $3s/(4\pi r)$ over the divisor E_g ; the clock's share $U = c/\Gamma$. Steps: $U = 3s/(4\pi r \Gamma E_g)$, and nature's $U = GM/(c^2 r)$ with $M = s$ in quanta and $c^2 = 1/3$ gives $G/c^2 = 3/(4\pi \Gamma E_g)$, that is $G_{\text{clock}} = 1/(4\pi \Gamma E_g)$; Kepler's G , read from the orbits of matter, is $[2 \cos \omega_0/(1 + \cos \omega_0)] G_{\text{clock}}$ by Eq. (14).